EXPERIMENT 5: AFFINE CAESAR CIPHER

1. Aim

To write and execute a C program to implement the affine Caesar cipher using the formula $C = (ap + b)\ 26$and determine the restrictions on the values of $a$and b.

2. Algorithm

1. Start the program.
2. Initialize the plaintext and key values $a$and b.
3. Check whether $a$is relatively prime to 26.
4. Calculate the ciphertext using $C = (ap + b)\ 26$.
5. Display the encrypted text.
6. Stop the program.

3. CODE

```
import subprocess
code = r'''
#include <stdio.h>
int main()
{
    char text[] = "HELLO";
    int a = 5, b = 8, i;

    printf("Plaintext: %s\n", text);
    printf("a = %d, b = %d\n", a, b);

    for (i = 0; text[i] != '\0'; i++)
    {
        text[i] = (a * (text[i] - 'A') + b) % 26 + 'A';
    }
    printf("Ciphertext: %s\n", text);
    return 0;
}
'''
with open("affine.c", "w") as f:
    f.write(code)

compile_result = subprocess.run(
    ["gcc", "affine.c", "-o", "affine"],
    capture_output=True, text=True
)

if compile_result.returncode != 0:
    print(compile_result.stderr)
```

```
with open("affine.c", "w") as f:
    f.write(code)

compile_result = subprocess.run(
    ["gcc", "affine.c", "-o", "affine"],
    capture_output=True, text=True
)

if compile_result.returncode != 0:
    print(compile_result.stderr)
else:
    result = subprocess.run(
        ["./affine"],
        capture_output=True, text=True
    )

    print(result.stdout)

    with open("affine_output.txt", "w") as f:
        f.write(result.stdout)
```

OUTPUT:

```
Plaintext: HELLO
a = 5, b = 8
Ciphertext: RCLLA
```

RESULT:

Thus, the C program for the affine Caesar cipher was implemented and executed successfully, encrypting the plaintext HELLO into the ciphertext RCLLA using $a = 5$and b=8.